

Practice Problems and Assignment

Lecture 1 and 2:

1. **(Mid)** Calculate the degree sum in a complete graph of 5 vertices. Derive a general formula for calculating the degree sum of a complete graph with “n” vertices.
2. **(Mid)** What is the highest size of clique you can get in W_8 ?
3. **(Mid)** Draw P_4 , C_5 and W_7
4. Prove that W_n doesn't have an induced K_4 for $n > 4$.
5. **(Mid)** Prove that number of odd degree vertices is even.
6. **(Mid)** Derive the formula for the number of edges for W_n , C_n , P_n .

Lecture 3:

1. **(Mid)** Prove that, “For any graph of seven vertices, G or G' contains an induced K_3 ”
2. **(Mid)** Prove that, “If a graph G is a complete graph, then it can not contain an induced P_3 ”

Lecture 4:

1. **(Mid)** 7,7,6,6,4,1 is this a valid degree sequence? Give proper reasoning.
2. **(Mid)** 5,5,3,1 Is this a valid degree sequence? If yes, then draw a graph (possibly a multigraph) with this degree sequence.
3. **(Mid)** Is 7,4,3,3,2,1,1,1 a graphic sequence? If yes, then draw a simple graph with this sequence.
4. **(Mid)** Show the adjacency matrix and adjacency list representation of the above graph.
5. **(Mid)** Suppose, you have a cycle graph of 6 vertices. Show the adjacency list representation of that graph.
6. **(Mid)** Let's say you have an adjacency list representation of a cycle graph C_n . What is the space complexity? What is the time complexity of checking an edge between two vertices u and v . Write the best possible answer.

Lecture 5:

1. **(Mid)** You have 6 vertices. What is the possible number of simple undirected graphs with those vertices? What is the possible number of simple directed graphs with those vertices?
2. **(Mid)** Let's say you want to add an additional constraint on problem 1. Among the vertices, the sixth vertex can only have outgoing edges. Then how many simple directed graphs are possible?
3. **(Mid)** Draw the complement graph of $K_{3 \times 4}$. What is the clique of the maximum size of that graph?

Lecture 6:

1. **(Mid)** Is it possible to have a path that is not a trail? Give proper reasoning.
2. **(Mid)** Let a graph G be a disconnected graph. Show that the complement of G is a connected graph.
3. **(Mid)** You have taught your friend about the lemma "*Every graph with n vertices and m edges has at least $n - m$ connected components*". Then your friend says the following, "*If I have a complete graph with 4 vertices then that graph has 6 edges. According to the lemma it means I have $4 - 6 = -2$ connected components which is clearly false, Thus the lemma is false*". Explain what is wrong with the above argument.
4. **(Mid)** What is the necessary and sufficient condition of the following statement "An edge is a cut-edge if and only if it belongs to no cycle". ?
5. **(Mid)** How many cut edges does P_n have?
6. **(Mid)** Prove that, "*If there is a circuit starting and ending with vertex "x" then there is a cycle starting and ending with vertex "x".*"
7. **(Mid)** Prove the necessary condition of, "*A graph G is bipartite if and only if G doesn't contain any odd cycle*".

Lecture 7, 8, 9:

1. **(Mid)** Write the necessary and sufficient statement of "*A connected graph G is Eulerian if and only if every vertex of G has even degree*".
2. **(Mid)** Prove that, "*A connected graph G is Eulerian if and only if every vertex of G has even degree*".
3. To practice the cycle decomposition of Eulerian graphs and finding Eulerian circuits in a Eulerian graph, just draw a graph with all even degree vertices. Then try to solve for that graph.

4. **(Mid)** Prove the sufficient condition of, "*A graph G is bipartite if and only if G doesn't contain any odd cycle*".

Lecture 10:

1. **(Mid)** For a complete graph K_n to be Hamiltonian and Eulerian, what is the condition that needs to be imposed on " n "? (A graph is hamiltonian if the graph has a hamiltonian cycle.)
2. **(Mid)** Find two edge disjoint hamiltonian cycles in K_5
3. **(Mid)** Prove that, "Let G be a connected simple graph. Then $\kappa'(G) \leq \delta(G)$ ".
4. **(Mid)** Find $\kappa(G)$, $\kappa'(G)$ and $\delta(G)$ of a wheel graph W_n .

Lecture 11:

1. Prove that, "Let G be a connected simple graph. Then $\kappa(G) \leq \delta(G)$ ".